would have to be the radius of the outside circle, 200/π kilometers, minus the radius of the horizon, 100/π kilometers; that is, it would have to be 200/π – 100/π = 16 kilometers (approximately). To accommodate the radial distance's far larger, 37-kilometer size, the surface must have the curved, trumpet-horn shape shown in the diagram

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Box 11.1

Perfection of Rulers and Clocks

By "perfect clocks" and "perfect rulers" I shall mean, in this book, clocks and rulers that are perfect in the sense that the world's best clock makers and ruler makers understand: Perfection is to be judged by comparison with the behaviors of atoms and molecules.

More specifically, perfect clocks must tick at a uniform rate when compared with the oscillations of atoms and molecules. The world's best atomic clocks are designed to do just that. Since the oscillations of atoms and molecules are controlled by what I called in earlier chapters the "rate of flow of time," this means that perfect clocks measure the "time" part of Einstein's curved spacetime.

The markings on perfect rulers must have uniform and standard spacings when compared to the wavelengths of the light emitted by atoms and molecules, for example, uniform spacings relative to the "21-centimeter-wavelength" light emitted by hydrogen molecules. This is equivalent to requiring that when one holds a ruler at some fixed, standard temperature (say, zero degrees Celsius), it contain always the same fixed number of atoms along its length between markings; and this, in turn, guarantees that perfect rulers measure the spatial lengths of Einstein's curved space-time.

The body of this chapter introduces the concept of "true" times and "true" lengths. These are not necessarily the times and lengths measured by perfect clocks and perfect rulers, that is, not necessarily the times and lengths based on atomic and molecular standards, that is, not necessarily the times and lengths embodied in Einstein's curved spacetime.
